Image Analysis/Classification

Typical Remote Sensing process

- Image data acquisition
- Importing images into image processing software
- Initial image statistics and display to assess image quality
- Preprocessing and Image restoration: radiometric and geometric corrections
- Image enhancement and transformation

Typical Remote Sensing process

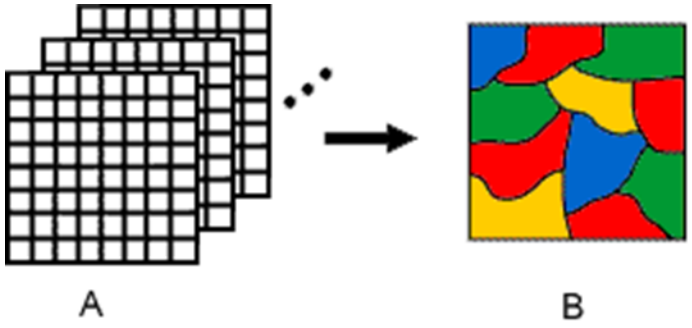
- **Image analysis for thematic information extraction (includes classification)**
- **Accuracy Assessment**
- **Applications: for example Change Detection**
- **Integration with other data in a Geographic Information System**

Image Classification

- A process of converting remotely sensed data into useful information
- An important part of image analysis is
  - Identifying groups of pixels that have similar spectral characteristics and
  - Determine the various features or land covers represented by these similar spectral groups

Image Classification

- Image classification can be described as a process to automatically assign land cover class categories to every pixel of the image
- A process of converting remotely sensed data into useful information



Visual Interpretation vs. Classification

- **Visual Interpretation**
  - **Human analyst attempts to classify features in an image using the elements of visual interpretation**

- **Digital image classification**
  - **Uses the spectral information represented by the digital numbers in one or more spectral bands, and attempts by using a computer, to classify each individual pixel based on this spectral information. This type of classification is also termed spectral pattern recognition**

Image Classification

- An image represents both spectral and information classes which must be distinguished while performing classification
- Spectral classes: groups of pixels that have nearly uniform spectral characteristics in all the spectral bands
- Information classes: various themes or object that represent the actual features on ground, *Classes* that human beings define

  e.g., Deciduous forest, Residential land, etc

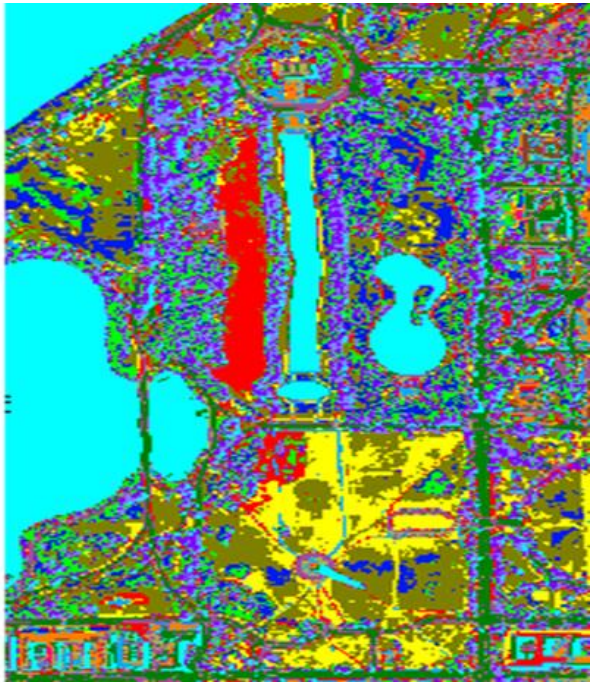
Information Classes and Spectral Classes

- A broad information class may contain several spectral sub-classes with unique spectral variations
- In Forest class , spectral sub-classes may be due to variations in age, species, and density, or perhaps as a result of shadowing or variations in scene illumination
- It is the analyst's job to decide on the utility of the different spectral classes and their correspondence to useful information classes

Spectral and Information Classes

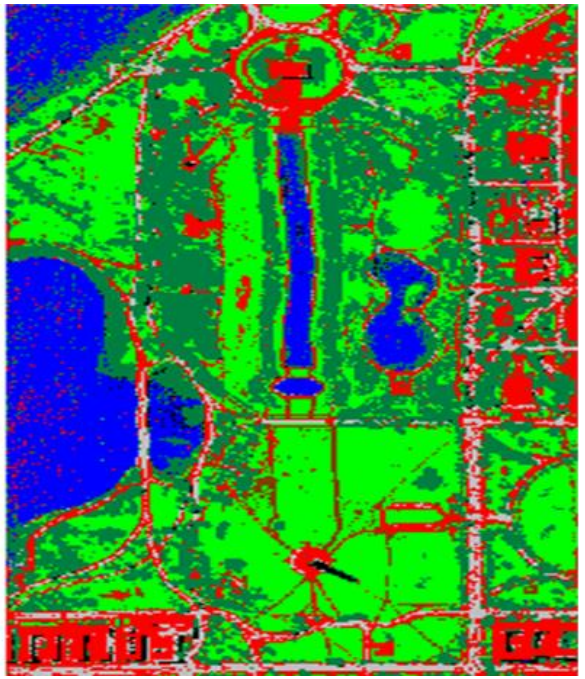


Image





Pixel groups with similar
Spectral Response



Classes

- Roof
- Street
- Path
- Grass
- Trees
- Water
- Shadow

Spectral and Information classes

- **Ideal classification results can be achieved once one spectral class is related to one information class.**
- **Likely results can be obtained once more than one spectral class relates to one information class.**
- **Problematic results: when spectral classes relate to more than one information classes, the results are problematic and likely inaccurate.**

Spectral Classes	Identity of Spectral Class	Corresponding Desired Information Category
Ideal Results	Water	Water
1	Coniferous trees	Coniferous trees
2	Broad Leave trees	Broad Leave trees
3	Bare Soil	Bare Soil
4	Rocks	Rocks
5	Built-up Area	Built-up Area
Likely Results		
1	Water	Water
2	Coniferous trees	Forest
3	Broad Leave trees	
4	Bare Soil	Open Land
5	Rocks	
6	Roads	Built-up Area
7	Urban Area	

Problematic Results	Spectral Class	Information Classes
1	water	Water
		Shadow
2	Bare land	Bare land
		Parking area

Classification

- Multispectral classification may be performed

- **Using**
  - Supervised or
  - Unsupervised classification logic
- **Using**
  - Hard or
  - Soft (fuzzy)
- **Using**
  - Per-pixel or
  - Object-oriented classification logic
- Hybrid approaches (using ancillary data/information)

Image Classification

- **Conventional classification approaches operate on a per-pixel basis**
  - **Per-pixel methods examine each pixel of the image independently and assign it to a class based on its spectral characteristics**
  - **The most used per-pixel methods are,**
    - **Supervised classifications**
    - **Unsupervised classifications**

Classification

- **Multispectral classification may be performed using**
- **Use of**
  - **Hard or**
  - **Soft (fuzzy)**
- **Use of**
  - **Per-pixel or**
  - **Object-oriented classification logic**
- **Hybrid approaches (using ancillary data/information)**

Classification

- **Parametric methods**
  - **assume normally distributed remote sensor data**
  - Require using sample data to make inferences about population parameters

- **Nonparametric methods**
  - **may be applied to remote sensor data that are not normally distributed**
  - **such as rule-based decision tree classifiers**

Per-pixel vs. Object-oriented Classification

- **Per-pixel classification**
  - **Digital image classification based on processing the entire scene pixel by pixel**
- **Object-oriented (based) classification**
  - **Techniques that allow the analyst to decompose the scene into many relatively homogenous image objects (segments) using a multi-resolution image segmentation process**
  - **The various statistical characteristics of these homogeneous image objects in the scene are then subjected to traditional statistical or fuzzy logic classification**
  - **Object-oriented classification based on image segmentation is often used for the analysis of high-spatial-resolution imagery**

Land-use and Land-cover Classification

- **Land use**
  - **refers to what people do on the land surface (e.g., agriculture, commerce, settlement)**
  - **relates to the human activity or economic functions associated with a specific piece of land, residential area ,industry**
- **Land cover**
  - **refers to the type of material present on the landscape (e.g., water, sand, crops, forest, human-made materials buildings, road)**

U.S. Geological Survey's *Land-Use/Land-Cover Classification System for Use with Remote Sensor Data*

- **The system is designed to be driven primarily by the interpretation of remote sensor data obtained at various scales and resolutions**
- **Anderson Paper uploaded on LMS**

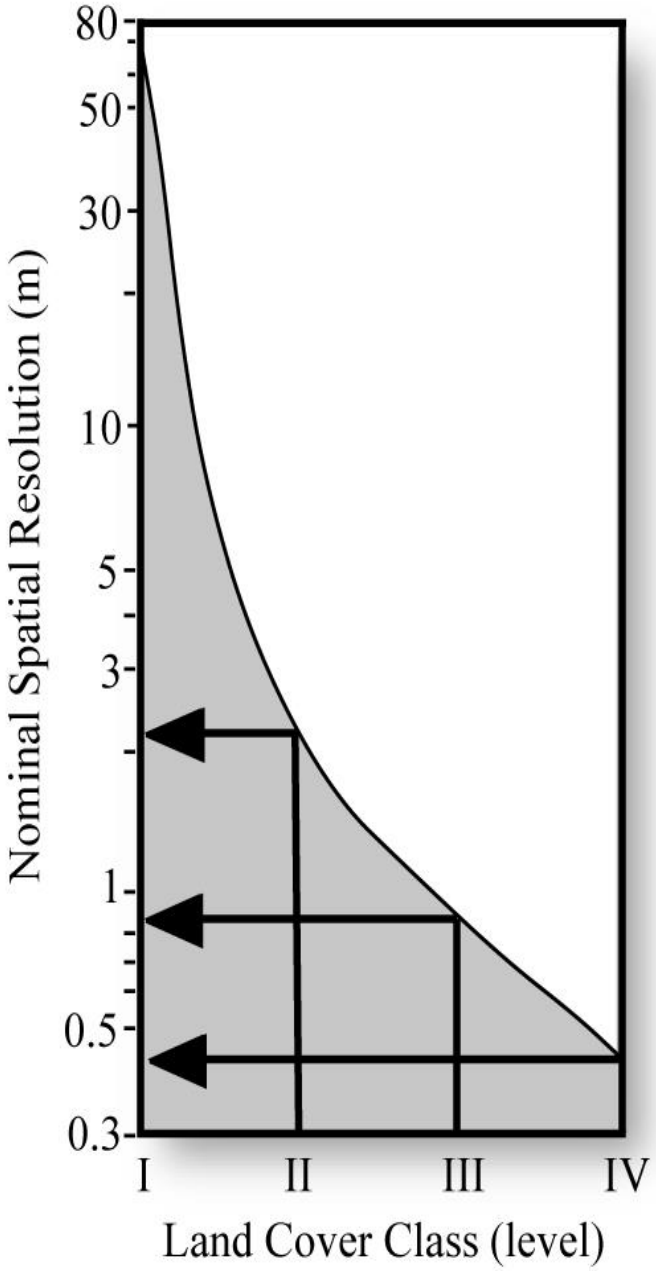
Level I	Level II
1. Urban or built-up land	11. Residential 12. Commercial and service 13. Industrial 14. Transportation, communications and utilities 15. Industrial and commercial complexes 16. Mixed urban or built-up land 17. Other urban or built-up land
2. Agricultural land	21. Cropland and pasture 22. Orchards, groves, vineyards, nurseries, and ornamental horticultural areas 23. Confined feeding operations 24. Other agricultural land
3. Rangeland	31. Herbaceous rangeland 32. Shrub and brush rangeland 33. Mixed rangeland
4. Forest land	41. Deciduous forest land 42. Evergreen forest land 43. Mixed forest land
5. Water	51. Streams and canals 52. Lakes 53. Reservoirs 54. Bays and estuaries
6. Wetland	61. Forested wetland 62. Non-forested wetland
7. Barren land	71. Dry salt flats 72. Beaches 73. Sandy area other than beaches 74. Bare exposed rock 75. Strip mines, quarries and gravel pits 76. Transitional areas 77. Mixed barren land
8. Tundra	81. Shrub and brush tundra 82. Herbaceous tundra 83. Bare ground tundra 84. Wet tundra 85. Mixed tundra
9. Perennial snow or ice	91. Perennial snow fields 92. Glaciers

Land Use/Land Cover Classification Level-III

Level I	Level II
1. Urban or built-up land	11. Residential 12. Commercial and service 13. Industrial 14. Transportation, communications and utilities 15. Industrial and commercial complexes 16. Mixed urban or built-up land 17. Other urban or built-up land

Residential

- Single Family
- Multifamily
- Apartments
- Residential Hostels
- Mobile Home Parks
- Others



Nominal spatial resolution requirements as a function of the mapping requirements for Levels I to IV land-cover classes in the United States (based on Anderson et al., 1976).

Note the dramatic increase in spatial resolution required to map Level II classes